RURAL & LANDSCAPE PLANNING

Energy and Resources

- 7th lecture -

Lecturer:
Dr Ana Peric Momcilovic

Lecture outline

Key topics:

- Introduction: resource extraction and rural studies
- Environmental justice framework
- Case study 1: Mineral extraction, Brazil
- Case study 2: Just transitions

Key takeaways

Introduction

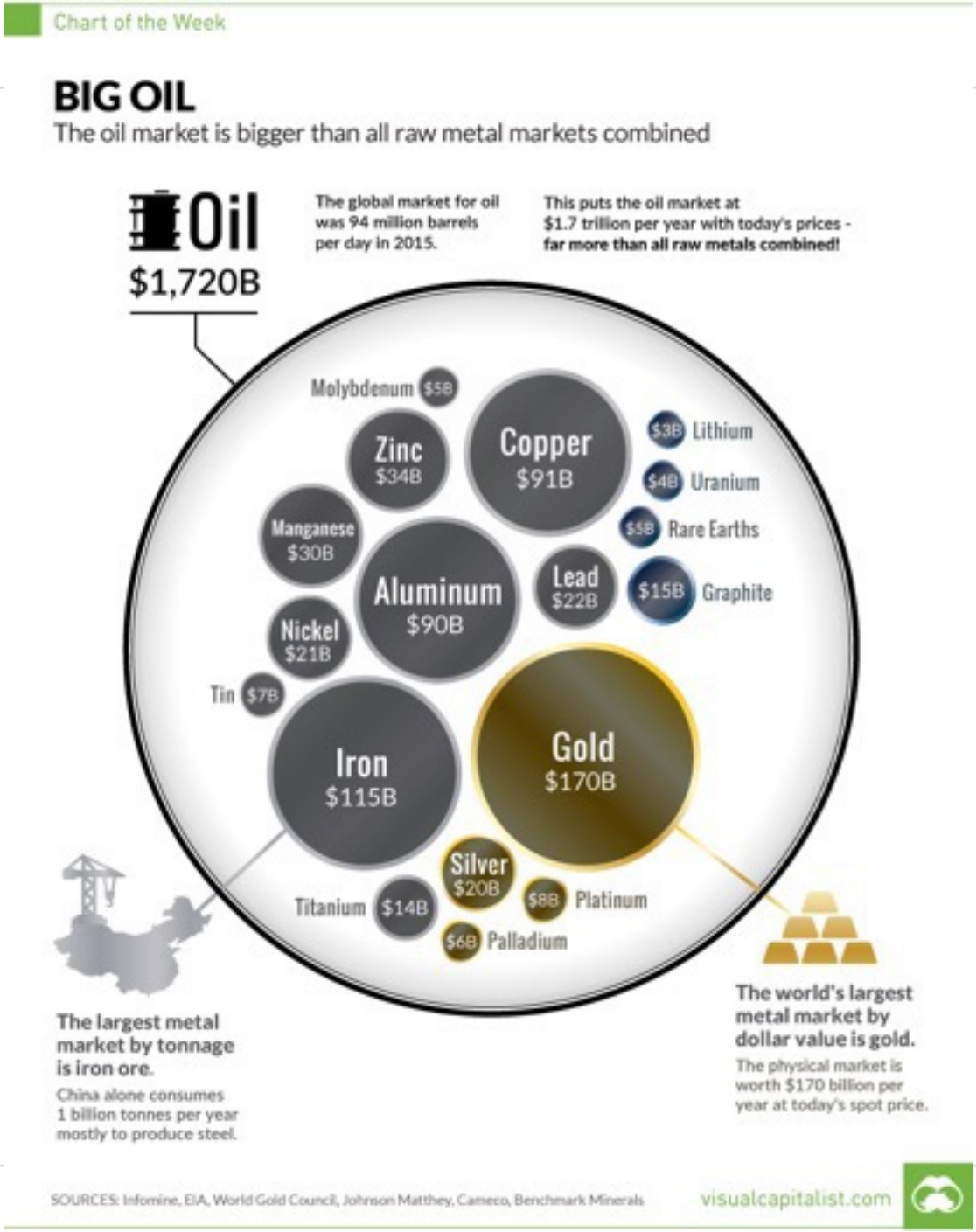
Mineral / resource extraction

- Part of the traditional backbone of the rural economy
- Rural as a resource sink – exploitation
- What does it mean for rural communities?

>> Example: extraction of lithium has numerous adverse effects on the catchment areas of extraction sites

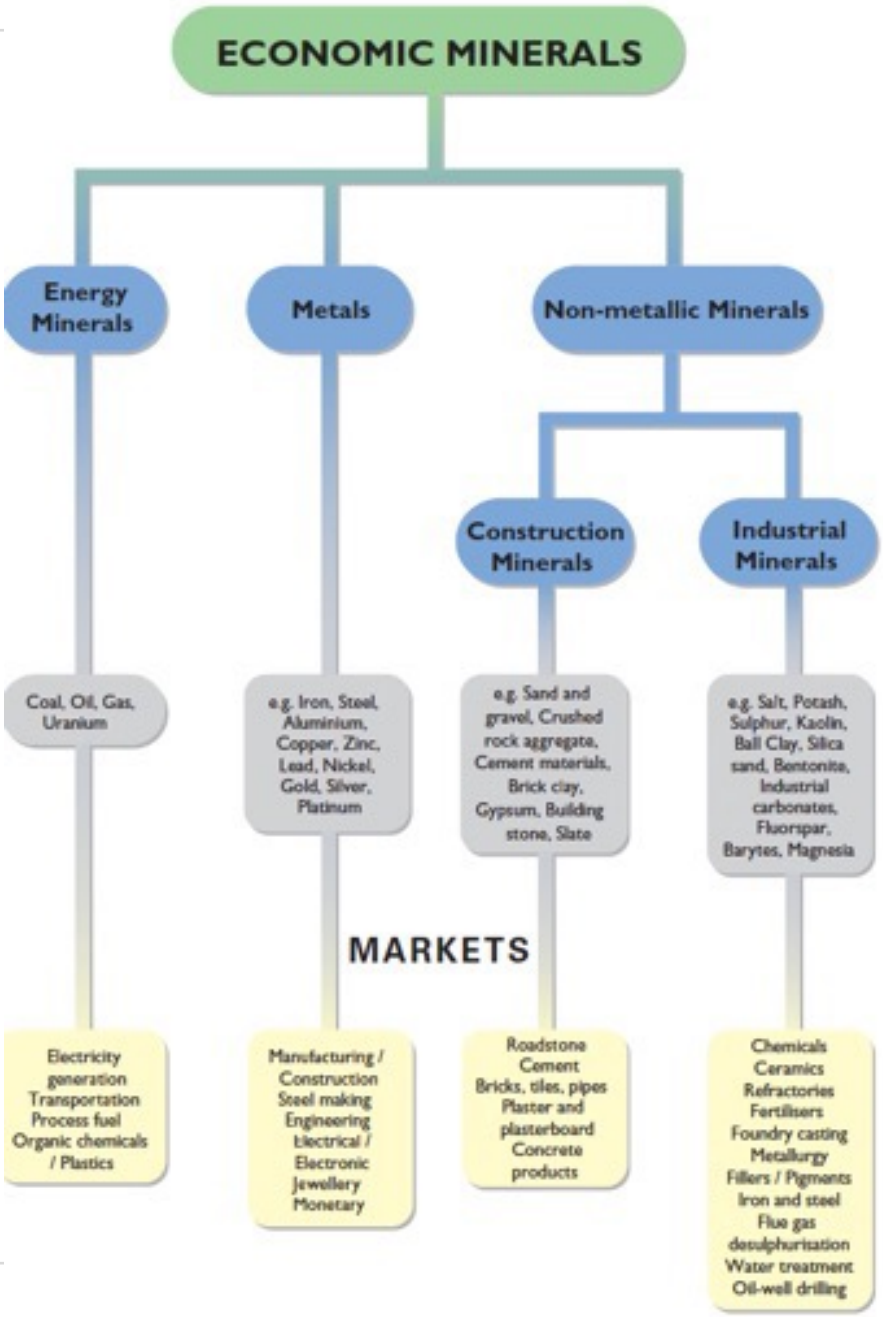
Introduction

- Global commodities market



Introduction

- Humans have exploited earth's resources throughout history
- Today, resource extraction underpins energy, manufacturing, construction, etc.



Introduction

Open Pit coal mining, Australia

Introduction

Tar Sands Oil Extraction, Alberta, CA

Introduction

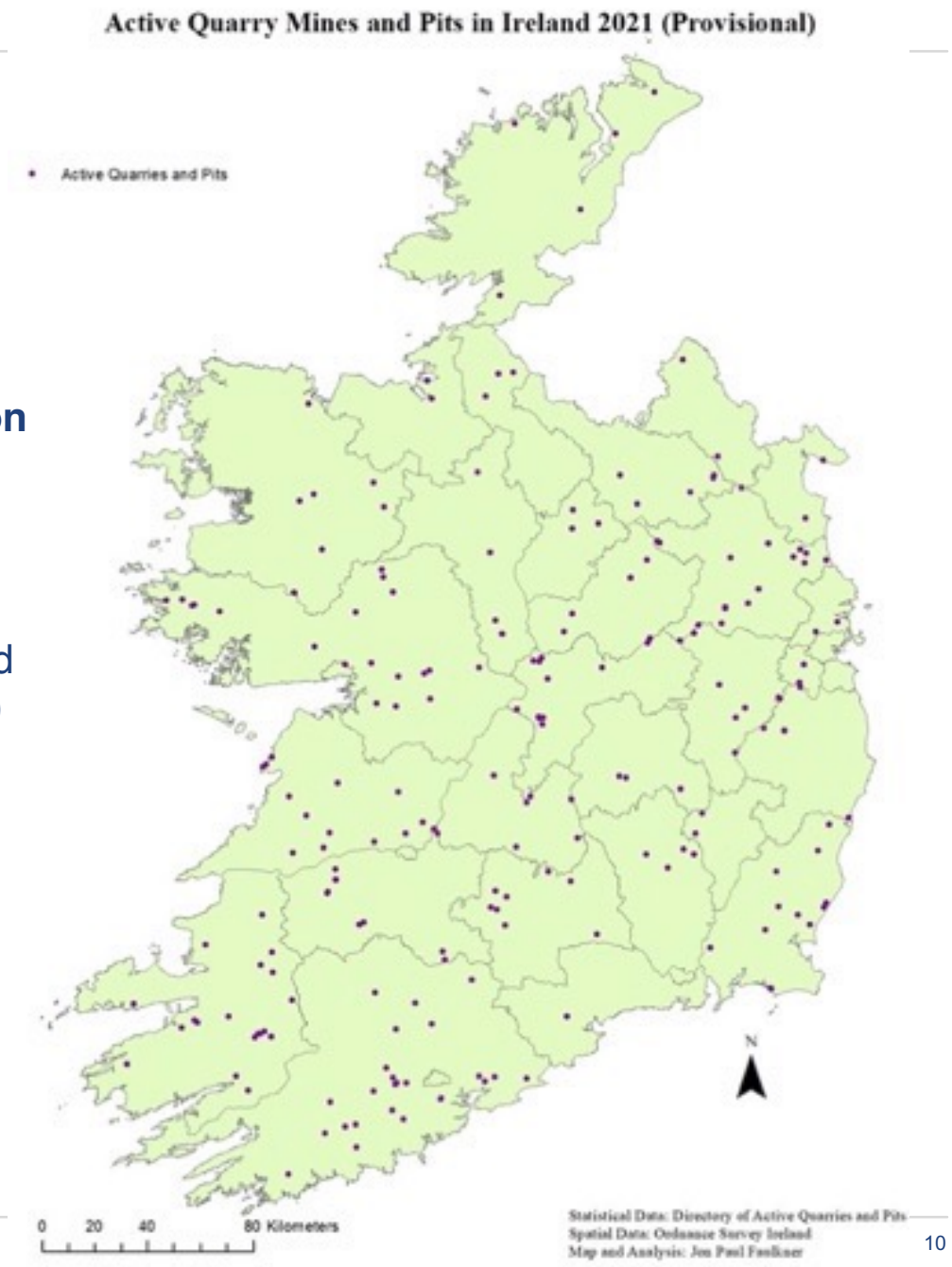
Aggregates Quarry, Co. Clare, Ireland

Introduction

Industrial harvesting of peatlands, Ireland

Extraction in Ireland

- Aggregates are essential raw materials for Ireland's **construction sector and infrastructural** needs
- Current demand for aggregates in Ireland is 12 tonnes per capita, double the EU-28 average demand (Irish Concreate Federation, 2019)
- Demand will increase due to forecasted demand for new residential and infrastructural development
- A total of 3,733 persons were employed in the mineral extraction sector in 2016 (CSO, 2016)



Extraction

Fracking –
natural gas exploitation from shale rock formations

Disadvantages:
- air and water pollution, soil contamination, habitat destruction
- respiratory problems
- earthquakes
- methane release

>> politically contentious in EU, widely accepted in US

Ireland joins France, Germany and Bulgaria in banning fracking

Private Members' Bill passed in Seanad after passage earlier through Dáil

Wed, Jun 28, 2017, 21:18

Marie O'Halloran

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An anti-fracking campaign group in jubilant mood after the fracking Bill was passed banning all fracking in Ireland. Photograph: Brenda Fitzsimons

The geopolitics of energy security

FINANCIAL TIMES

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Your guide to a disrupted world

UK energy + Add to myFT

Boris Johnson opens door to fracking in response to Ukraine war

Prime minister looking at 'all options' as he prepares to draw up new energy strategy

Ireland can play part in ending dependence on Russian fuel

Hannah Daly, Paul Deane, Olexandr Balyk and Diarmuid Torney
Opinion

Ukraine crisis provides chance to accelerate sustainable energy transition

> "In solidarity with the people of Ukraine, individuals, communities, and businesses can play their part in a collective effort to quickly reduce fossil fuel consumption"

Rural studies and extraction

- Longstanding tradition of examining the relationship between resource extraction and rural communities (since the 1940s)

>> Studies often focus on **'boom + bust'** nature of relationship

>> resource-dependent communities and market volatility
>> single-purpose communities

>> **Conflicts with other rural sectors** (e.g., farming, tourism)

Rural studies and extraction: key topics in the literature

Resource boomtowns	Examining the impacts on remote settlements of rapid growth in terms of benefits to local economies (e.g. Measham and Fleming, 2014), shifting community relations (e.g. loss of community control, e.g. Miller et al., 2012) and decline following resource depletion. Much attention focused on the long term vulnerability of resource-dependent localities (e.g. Tonts et al., 2012)
The 'resource curse'	Exploring and explaining the negative impact of natural resource abundance on long-term economic growth at macro scale (e.g. Auty, 1993) Increase in studies focused on the local scale resource-curse, and impacts of mining on local livelihoods, poverty and environmental degradation (e.g. Papyrakis, 2017; Hilson, 2010)

Rural studies and extraction: key topics in the literature

Indigenous peoples' rights	Investigating conflicts between mining operations and indigenous peoples' livelihoods (often based on land resources) (e.g. Hilson, 2002) Methods to address impacts on indigenous communities through so-called mining payments, employment schemes, and impact-benefit agreements (e.g. O'Faircheallaigh, 2004, 2013; Caine and Krogman, 2010)
Post-mining landscapes and economies	Examining the environmental degradation of mining operations and landscape rehabilitation following closure of mining works (e.g. Collier and Scott, 2008, 2009, 2010). Exploring the search for a post-mining viable economy (Neil et al., 1992; McDonald et al., 2012) and transitions to more sustainable economic pathways (e.g. Fazito et al., 2016)

Rural studies and extraction: key topics in the literature

From local to global environmental opposition	Examining local environmental degradation resulting from mining operations (e.g. Conde, 2017) Upscaling of environmental concerns to global issues, particularly related to carbon intensive activities and climate change (e.g. coal or new fracking operations) (e.g. Connor et al., 2009)
New governance modes and power	Charting new modes of governance between mining companies and local communities, including the formation of new private-community partnerships (e.g. Cheshire et al., 2014) and models of corporate responsibility (e.g. Cheshire, 2009) Models of decentralised decision-making, but also highlighting the emergence of conflicts among local elites (e.g. Arellano-Yanguas, 2011)

Environmental justice

- In recent years, it has become increasingly apparent that the issue of environmental quality is inextricably linked to that of human equality
- Whenever in the world there is environmental degradation/despoilation, it is almost always linked to questions of social justice, equity, rights and people's quality of life

Environmental justice

Two dimensions to this (Agyeman et al., 2003):

>> Globally, countries with a more equal income distribution, greater civil liberties and political rights tend to have higher environmental quality (e.g., air and water pollution, access to clean water)

>> Environmental problems bear down disproportionally upon the poor – both within countries and globally

Environmental justice is…

The linking of environmental issues to concerns with social justice

Environmentalism of the poor (Martinez-Alier, 2002)

>> A realisation that a sustainable society is a just society – locally, nationally and internationally, and both within and between generations and species

Origins of environmental justice movement

- A longstanding tendency to locate sewage, municipal waste, etc. near to where the poor, minorities, and politically disempowered since ancient Greece and Rome
- Facilities are located in these communities as they offer 'the path of least resistance'
- As a social movement, environmental justice has its origins in USA
- Traced to citizen revolts against the siting of toxic waste and hazardous and polluting industries in areas inhabited by poor and ethnic minority populations

>> Anti Toxics Movement (e.g., the Center for Health, Environment & Justice (USA)), 1970s: tackling environmental disasters to eliminate the exposure to toxics in the environment >> env. regulations

>> Environmental Racism: disproportionate burden of env. hazards on communities of colour

A brief history: US

Key principles

Early action focused on a ***distributional concept*** of justice

- Based on John Rawls' classic 'A Theory of Justice'

>> also known as 'Justice as Fairness': framework for a just and fair society

>> '**original position'**, individuals placed behind a **'veil of ignorance':** this prevents them from knowing their social status, abilities and personal preferences

>> two principles of justice:
- equal basic liberties (speech, assembly, religion)
- difference: economic inequalities allowed as long as they benefit the least advantaged members of society

Key principles

Early action focused on a ***distributional concept*** of justice

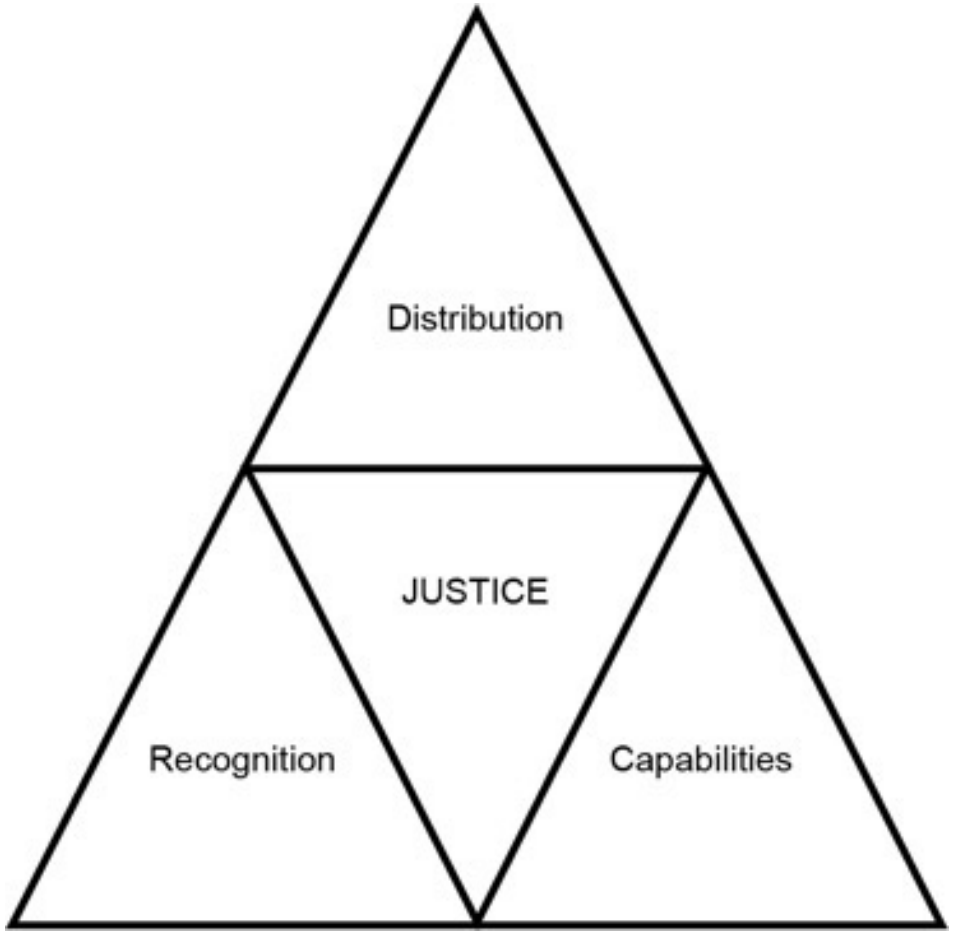
- Based on John Rawls' classic 'A Theory of Justice'

>> A focus on **'fairness' in the rules** that govern the just distribution of goods (*procedures*)

>> Every locality should have an **equal share of environmental 'goods' and 'bads'**, regardless of the race or class of its population (*outcomes*)

>> How and what gets distributed

Towards an environmental justice framework



David Schlosberg (2007)

Environmental justice framework: Distribution

- Outcomes: *what* gets distributed
- Procedure: *how* goods and bads gets distributed

Participation is key

>> Procedural fairness
>> Inclusion
>> Empowerment
>> Giving people voice
>> Confronting power and oppression

Environmental justice framework: Recognition

- Examines the **context of oppression**, which are the **causes of maldistribution**
- Caused by a lack of recognition in the **social and political realm**
    - Demonstrated by **forms of insults**, degradation, and devaluation at both individual and cultural level
    - This inflicts **damage to oppressed** individuals and communities
    - **Injustice as the outcome** of various social and institutional relations
    - Recognition is key to self-worth and human dignity, but undermined by:
        - Disrespect, denial of rights and denigration of ways of life
        - Forms of institutional practice

Environmental justice framework: Capabilites

'Development as Freedom' – Amartya Sen (1999)

- Moving beyond simple distribution concerns, BUT **how distribution of goods and bads affect our well-being** and how we 'function'
- Capabilities are about a **person's opportunities** to do and to be what they choose in the context of a given society
- Focus on **individual agency**, functioning and well-being
- Capabilities – those things that enable us to **translate basic goods into the functioning of human life**

Towards an environmental justice framework

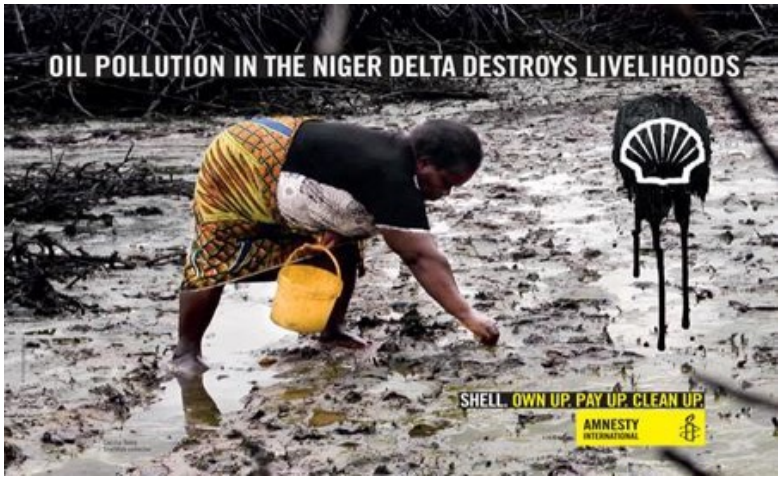
Towards a **transnational** approach

- In the context of globalisation and multinational corporations, 'environmental bads' are now displaced across borders to countries with lower environmental regulation and standards

- Companies usually locate their dirtiest businesses in areas that offer the least political resistance

  >> Example: Niger Delta and oil in Nigeria
  >> Example: the exploitation of minerals in Brazil

Towards an environmental justice framework



Case study 1:

Mineral extraction conflicts in the Espinhaco Range Biosphere Reserve, Brazil

Case study 1:

Background:

- Historic mining area
- Recognised as a UN Biosphere Reserve in 1990s
- Promotion of sustainable tourism (late 1990s-2000s)
- Boom in minerals – mining once again profitable

Case study 1: Brazil

Distributional justice:

- Global mining company vs. fragmented local environmental movements
- Use of shell companies to purchase land rights
- Exploited local lack of knowledge in acquiring land rights + individual negotiations (to avoid compensation)

Case study 1: Brazil

Recognition:

- Local values, knowledge and experiences were ignored

  >> Exclusion

- Policy reduced to technical appraisals

  >> Favours well-resourced parties

- Fostered local political divisions
- Political corruption

Case study 1: Brazil

Capabilities:

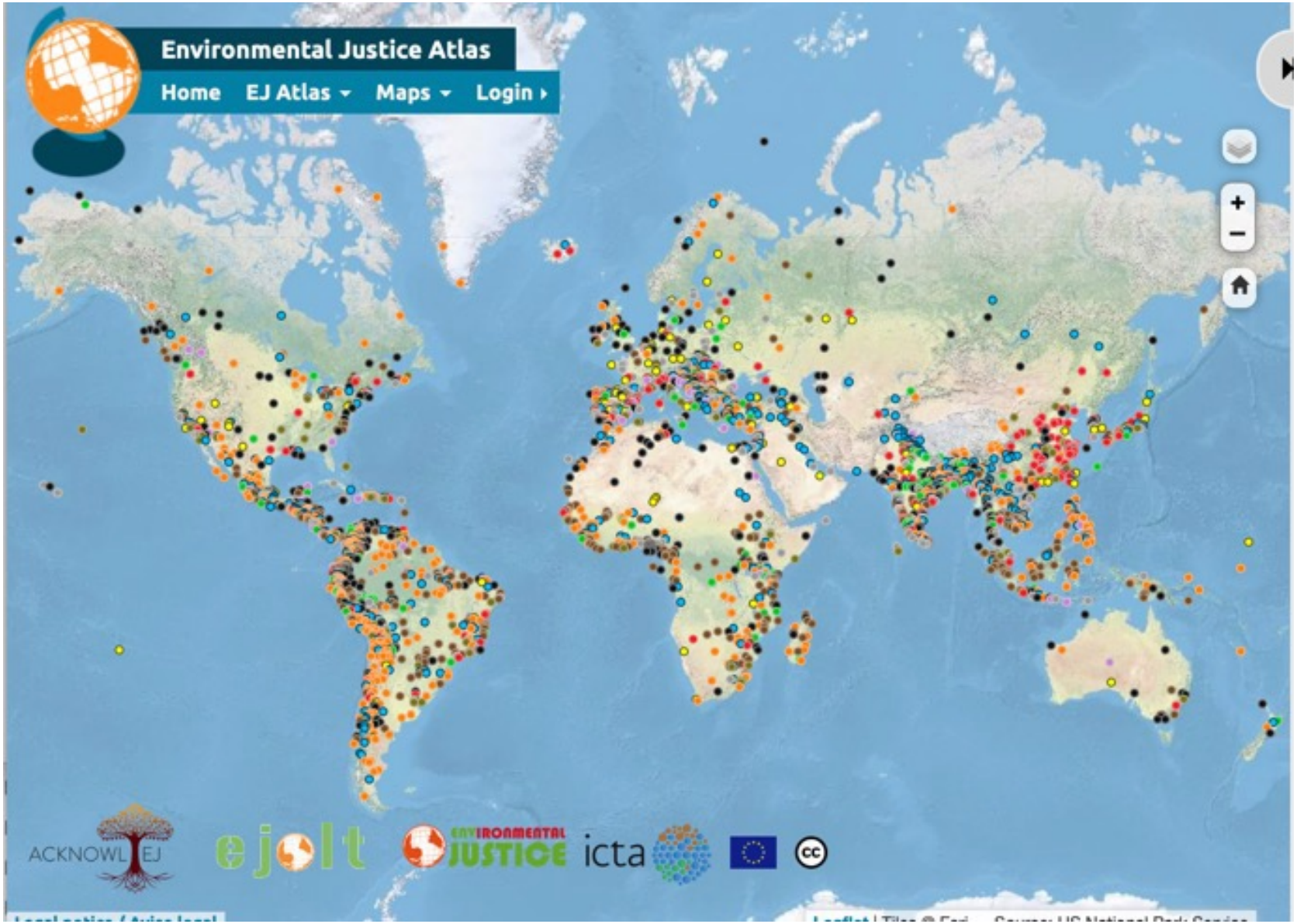
- Opposition grew to mining due to impacts on local livelihoods and wellbeing
- Main opposition not to the mine (its distribution) but to erosion of wellbeing (capabilities)
- Exclusion from traditional land
- Degradation of local springs and water quality
- Impacts on farming
- Human rights violations

Case study 1: Brazil

Opposition and reclaiming rights:

1. Right to information and participation;
2. Right to freedom of meeting, association and speech;
3. Right to work and to a dignified standard of living;
4. Right to adequate housing;
5. Right to education;
6. Right to health and a healthy environment;
7. Right to a fair negotiation, isonomy, according to transparently and collectively constructed criteria;
8. Right to come and go;
9. Right to the traditional practices and life-style as well as access and preservation of cultural, material and immaterial heritage;
10. Rights of the indigenous people, *quilombola* (communities formed by runaway slaves during slavery in Brazil) and traditional communities;
11. Right of vulnerable groups to special protection;
12. Right to access to Judge and to a reasonable duration of the judiciary trial; and
13. Right to protect the family and the social and community solidary linkages.

MAPPING THE WORLD'S MINING CONFLICTS
Barrick Gold's plans for one of the largest gold mines in the US has been bitterly opposed by the Western Shoshone peoples who fear that toxic mine waste will affect water supplies and claim that their sacred mountain and culture would be irreparably damaged.
Chirano Gold Mine, a combined open pit and underground gold mine, is owned by Canadian company Kinross (acquired from Red Back Mining Inc in 2010). It has been marked by a long running compensation dispute, dating from 2004. Claims of pollution and human rights abuses have also been made.
The Tampakan Copper–Gold Project in the Philippines is one of Asia's largest copper–gold mines and may force 5,000 people to move. Because of opposition by the Bla'an and other tribes, the army and paramilitary groups have been deployed to defend the $2.5bn investment. This militarisation of the area has resulted in the deaths of many anti–mining and indigenous peoples' leaders.
Mined commodity key:
Gold
Uranium
Copper
Iron ore
Silver
CANADA
UNITED STATES
MEXICO
HONDURAS
GUATEMALA
NICARAGUA
PANAMA
COLOMBIA
ECUADOR
PERU
BRAZIL
CHILE
ARGENTINA
PACIFIC OCEAN
ATLANTIC OCEAN
SPAIN
PORTUGAL
HUNGARY
ROMANIA
GREECE
TURKEY
NIGER
GHANA
ZAMBIA
SOUTH AFRICA
INDIA
INDIAN OCEAN
CHINA
SOUTH KOREA
PHILIPPINES
INDONESIA
The Escondida copper mine in Chile is the largest in the world and employs 2,000 people, but it has been beset for many years by strikes and disputes over pay, working conditions and workers' health protection. It has destroyed a vast area of the Atacama desert.
Freeport's vast mining operation in West Papua, which has included the world's largest gold and third largest copper mine, has been contested for 40 years for its pollution and environmental devastation. In 1977 the rebel Free Papua Movement attacked the mine. In return the Indonesian military killed hundreds of people. Accusations of cultural genocide are denied by Freeport.
SOURCE: EJATLAS.ORG GRAPHIC: PETE GUEST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Case study 2: Decarbonising and just transitions

Closure of peat-fired power stations, Ireland

By Cianan Brennan

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Friday, November 08, 2019 - 04:24 PM

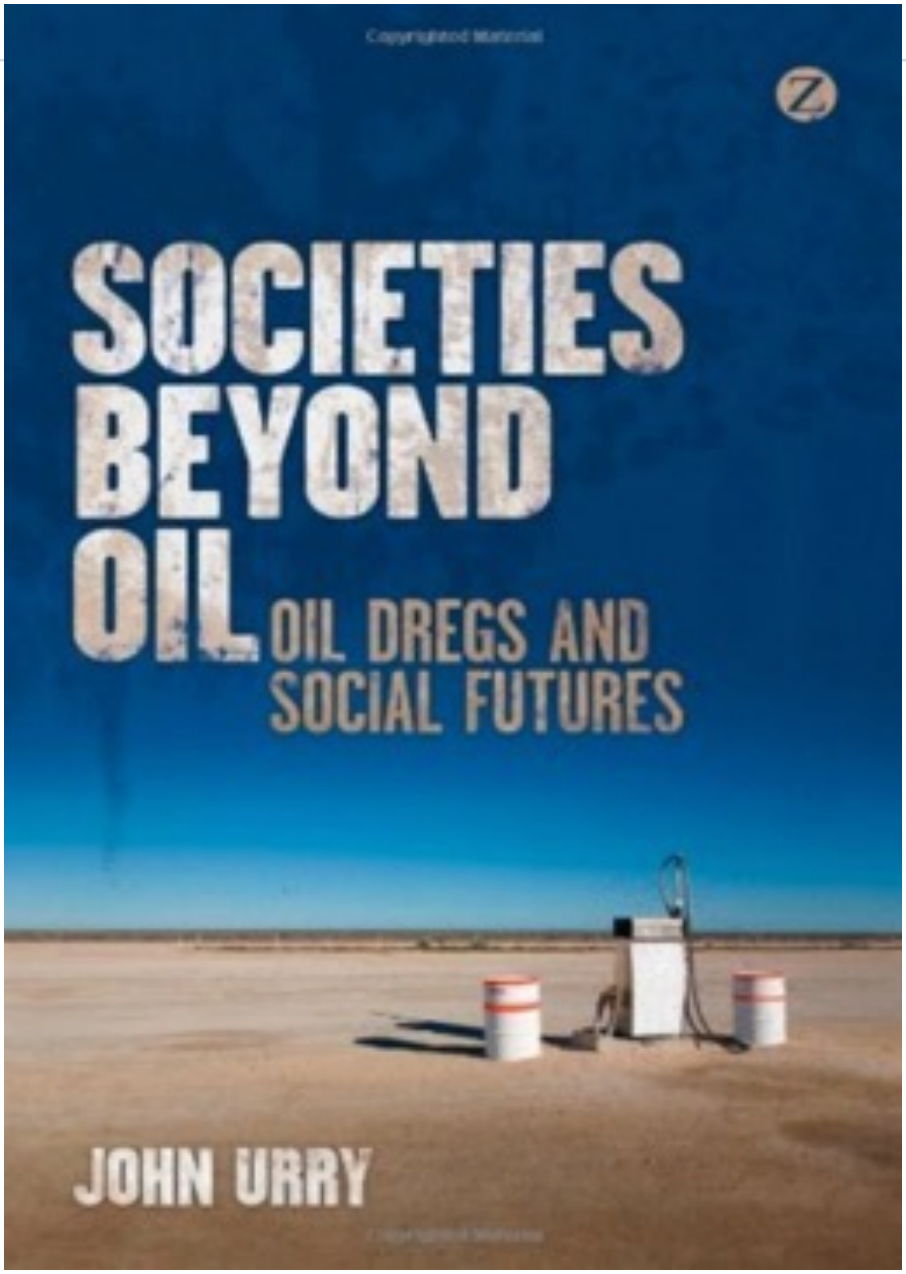
The Government has sought to allay disquiet in the midlands over the closure of two peat-fired power stations by appointing its first Just Transition Commissioner, Kieran Mulvey.

Solar energy farm and environmental degradation, Vietnam

- Soil and water quality
- Visual impacts
- Hazardous materials
- Heat island effect

Decarbonising and rural

- Oil fuels almost all movement of people and objects
- Oil is central in manufacturing, agriculture, and the distribution of water
- Decarbonisation requires systemic change



Decarbonising and rural

- Rural is both a site of energy production and a location of social resistance to low-carbon transitions
  (Phillips & Dickie, 2019)

- How can we ensure a just transition in rural places?

  >> E.g., impacts of carbon tax on dispersed rural communities

  - increased fuel costs
  - high energy costs (for electricity from fossil fuels)
  - impact on agriculture: reduced profitability

Justice framework

Distributional justice

>> Addressing the 'who gains' question

- Community benefits – e.g., community shares option
- Compensation – e.g., individual level, community-based
- Community ownership or assets-based approach
- Local employment and/or benefits to local businesses
- Design mitigation (siting, landscaping)
- Linked with environmental facilities

>> Global climate justice!

Justice framework

Procedural justice

- Independent advice on compensation – transparent negotiation with energy companies
- Plan-led approach – specific planning policies at regional/local scales
- Transparent planning process and decisions – fairness
- Pre-planning community engagement
- Avoid use of 'NIMBY' label – communities have valid concerns (recognition)

Case study: Midlands and a Just Transition

- Midlands region **dependent on resource extraction** and carbon intensive industry – peat-harvesting and peat-fuelled power stations
- **Environmental degradation**: biodiversity loss, scarred landscapes, carbon intensive land use
- Implications of decarbonisation: **immediate job losses** with ceasing of extraction and closure of power stations plus impact on local service economy
- Securing public and political support

Case study: Midlands and a Just Transition

Response:

- Just Transition Fund – community projects
- Energy transitions – home energy efficiency roll out
- Participation in EU schemes
- Bord na Mona's Brown to Green Strategy
  - Ceasing peat harvesting
  - Bog rehabilitation (employing existing employees involved in extraction)
  - Carbon sequestration
  - Recycling
  - Renewable energy sources

Key takeaways

- The relationship between resource extraction and rural communities
- Environmental justice as a framework for understanding (and overcoming) conflicts
- Just transitions